Tutorial XIV - Recap and Discussion

Applied Optimization with Julia

Introduction

This will (hopefully) be the easiest tutorial of the entire course. Essentially, we will just use today's tutorial as a last chance for you to ask questions and discuss the content of the course.

Recap

To structure the discussion (and your exam preparation), here is a compact overview of the problems we have modeled and the key modeling idea each one introduced:

Topic	Original Problem	Key Modeling Idea
Solar Panel Transport	Classic Transport Problem	Sets, parameters and continuous variables
Beer Production	Capacitated Lot-Sizing Problem (CLSP)	Binary setup variables and Big-M constraints
Split Order Minimization	Quadratic Multiple Knapsack Problem (QMKP)	Quadratic objective with binary variables
Library Routing	Capacitated Vehicle Routing Problem (CVRP)	Subtour elimination and heuristics
Police Service Districting	p-Median Problem	Contiguity and compactness constraints
Safety Planning for the Hajj Pilgrimage	Scheduling Problem	Time-indexed scheduling with penalties
Arena Seating	2D-Knapsack Problem	One binary variable per seating group
Passenger Flow Control	Dynamic Network Flow Problem	Queue dynamics over time periods

For the Julia and JuMP syntax, take a look at the Julia cheatsheet and the JuMP cheatsheet.

Some questions to test yourself with and to discuss in the session:

- Which of our problems needed binary variables, and why?
- Where did we use Big-M constraints, and what is the role of the “M”?
- Why do subtours appear in the routing model, and how did we prevent them?
- Which models were solvable exactly, and where did we need heuristics?
- If you had to simplify a real-world problem into a model, which assumptions would you question first?

> **Note**
> Exam logistics in short: you can bring **one DIN A4 sheet** with notes on both sides, and your bonus points count if your mark is at least 4.0. See the FAQ for the details.

Feedback on the lecture series

To receive half a bonus point, you will just need to complete a short feedback form on the lecture series before the exam. It is not from the university, but from me, and I would be truly grateful for your honest opinion and feedback. Your feedback is important to me and has absolutely no influence on your grade! The half point counts toward your exam bonus points, just like the tutorial submissions. In total I think it will take you about 15–20 minutes to complete the form - that’s not too bad given that you will receive half a bonus point for it, right?

In all honesty, I know this is a lot to ask of you - especially given that the exams are coming up. However, I think it is very important for me to get your honest opinion and feedback. I will use it to improve the course for future students, as well as for my future teaching.

If the form does not load, you can open it directly: https://tally.so/r/Ekl9gN. Note that the form is provided by Tally, a third-party service that processes your responses.

Wrap-up

Thanks for your participation! I wish you all the best for your exam and I hope you enjoyed the course! If you have any questions, please feel free to ask me or send me an email. We offer seminars each summer semester where you can explore different topics in more depth. In case you are interested in writing your thesis with me, please let me know and we can discuss potential topics.

Bibliography